DECISION RECORD

Fund evidence, retain the stop option

Conditional approval of up to **$1.50M for 90-day joint diligence**. Hold chronic tonsillar human use. Do not progress the proposed rigid tonsillar Utah-array design. Later capital requires a new release decision.

Boundary	Decision	What funding can establish
A. Hearing core	Conditional R&D;	Partner access, exact assembly, calibrated bench performance and a justified translational programme.
B. External assistant	Conditional prototype	Patient-controlled communication and listening support compared with current practical alternatives.
C. Pharyngeal sensing	Hold; rigid route no-go	Signal-origin and incremental-benefit tests first. A conformable cuff is a hypothesis, not a validated substitute.
Whole programme / software	No unrestricted financing	Separate responsibility, rights, approved configuration and support liabilities before commercial commitment.

Primary S2 financial scenarios - not probabilities

Metric	Whole-program base	Licensed software base
10-year NPV at 15%	-$95.53M	-$2.05M
Peak cumulative cash need	$197.15M	$8.63M
2036 revenue	$168.05M	$20.95M

> This hypothetical product assessment does not establish an assembled system, interoperability, operation in any identified person, a partnership, a licence or a marketing authorization.

S1: Due Diligence pp.1-7; S2: Proposal pp.1-4,19-26; financial_comparison.json; editorial_controls.json. Real 2026 USD; year-end 2027-2036 cash flows, zero terminal value.

INTEGRATION METHOD

Keep the evidence; resolve the conflicts

Use S1 for corrected scientific and privacy boundaries, S2 for the primary investment model and new controlled research extensions, and Gemini only for independently justified additions. Agreement among desk reviews does not reduce empirical uncertainty.

Stream / exact input	Retain	Modify or reject
S1 Neural_Audio_Due_Diligence (1).pdf; companion model	Component-specific evidence; cortical/peripheral separation; privacy limits; independent gates.	Financial schedules remain a visible alternative. Do not continue fixed later-stage ceilings or treat prior headlines as current primary values.
S2 UmboMic_Investment_Proposal.pdf; UmboMic_Valuation_Model.xlsx	Primary reconciled annuals; partner-development boundary; AttentivU, read-only controller and voluntary evaluation.	Retain unverified rights and approved-platform premises. No clinical benefit, attention treatment or diagnostic claim from research tools.
Gemini GEMINIDR_Neural-Audio CoPilot Platform.md	Candidate soak, anatomy, failure-injection and comparator tests; explicit owners.	Reject claimed achieved latency, validated cuff blanking, mandatory carotid margin, forced diagnostic causes and unreconciled financial rows.

Provenance rules

Rule	Application
Namespace original codes	S1:SCL002 and S2:SCL002 are explicit input references; Gemini section/line references remain separate.
Deduplicate by DOI / URL	Repeated papers and participants count once. A correction, figure dataset or follow-up report is not independent efficacy replication.
Label every number	Observed result, scenario calculation, or proposed requirement. Never convert a planning value into an achieved specification.
Bound the review	Primary sources selectively rechecked through 6 November 2026; no new laboratory experiments, systematic meta-analysis or sponsor-dossier audit.
Archive coverage	The PDF has 32 selected decision references. The full S1 ledger and 141 source-attributed S2 bibliography entries are retained in the master workbook archive; archival inclusion is not independent verification.

Source locations and 19 science dispositions: science_review.md. Financial sources and 93 original-cache checks: financial_comparison.json. Full S2 archive: stream2_full_references.json. R01-R32 are selected, deduplicated decision sources; the register is not exhaustive.

A promising component, not a validated TICI

The microphone evidence supports further engineering. The new hearing system still needs integrated acoustic, mechanical, electrical and chronic validation; essential hearing performance cannot be inferred from sensor noise alone.

Evidence	Established result	Transfer limit
Yeiser 2024 [R01]	Initial human cadaveric-ear proof; 32.3 dB SPL equivalent input noise (100 Hz-7 kHz).	No chronic integrated living-human performance.
Wawrzynek 2026 [R02]	Seven sensors / five human cadaveric ears; 32.4 dB EIN. Sensing region 3 x 3 mm; completed sensor about 20 x 4 mm.	Sensing-tip dimensions are not total implanted-system dimensions.
Calibration [R02]	58 fC/Pa at 1 kHz and 230 at 2 kHz include estimated outer-ear gain; 1 kHz without it is 26.	Reference plane and ear coupling must accompany sensitivity targets.
Fabrication [R02]	50 um PVDF layers and 15 um Parylene C. Accelerated aging remained future work.	Gemini 28 um layers and completed saline-soak evidence are rejected.
Comens 2026 [R03]	13 temporal bones / seven female sheep; functional tests in two specimens. Printed stainless-steel fixation.	Cadaveric study; human fixation under development. September correction only fixes name presentation.

> Retain a candidate accelerated-aging workstream. Validate failure mechanisms and acceleration assumptions before lifetime extrapolation; do not claim that a 67 C soak establishes years of service.

Keep: S1 pp.3-4; S2 pp.2,27. Correct: Gemini §§1.1,1.3 WS1,2.1-2.2,6.2. R01-R03. No measured integrated <2-5 ms hearing latency is supplied.

Results transfer only within their tested scope

Cortical speech results establish important capability in selected participants. They do not demonstrate speech decoding from a tonsillar cuff, unrestricted thought access or benefit for every cochlear-implant recipient.

Study family	Recorded modality / result	Required interpretation
Willett 2023 [R04]	Intracortical, one ALS participant; 62 words/min; 23.8% WER at 125,000-word vocabulary.	Task and vocabulary constrain the metric.
Metzger 2023 [R05]	High-density surface ECoG; one participant with paralysis; text, speech and avatar.	Not intracortical single units or a 256-channel Area 6v-only study.
Card 2024 [R06]	T15; four arrays / 256 electrodes; 2.5% final copy WER; 3.7% in 925 selected labeled conversation sentences.	Not every all-day conversation. Arrays were not exclusively in Area 6v.
Wairagkar 2025 [R07]	Voice synthesis using T15, also studied by Card.	One participant family; frame inference is not total playback latency.
Kunz 2025 [R08]	Task-dependent inner speech and gating experiments in four implanted participants.	A learned keyword is not the sole physical permission boundary.
Littlejohn 2025 [R09]	Streaming output increments of 80 ms; general-set median cue-to-speech 1.67 s, cue-to-text 1.56 s.	Separate chunk interval, intended onset and output latency.

> Reject Gemini's universal "256 Area 6v channels are necessary" explanation. Retain S1's modality distinction and S2's leakage-resistant calibration protocol. An adjacent signal is neither equivalent nor proven incapable.

S1 p.3; S2 pp.2,26; Gemini §§1.2,2.2,6.2. R04-R09. Severe speech/motor-impairment cohorts do not automatically define the target CI subgroup.

Anatomical access is not semantic access

The proposed pharyngeal route lacks target-specific chronic benefit evidence. Anatomical proximity, healed tonsillectomy tissue and transoral surgical access cannot establish a stable speech interface.

Candidate / claim	Decision	Evidence needed
Rigid penetrating tonsillar array	No-go for this proposed chronic design	Mechanical, mucosal, infection, vascular and nerve hazards without established benefit. This is a design/investment decision, not a measured complication rate.
Conformable submucosal mesh	Hold	Contact physics, tissue barrier, fixation, migration, selectivity, explantability and incremental utility. Flexibility is not proof of safety.
Named peripheral-nerve cuff	Hold	Define the actual nerve and fiber target. Separate ENG from muscle and motion contamination; measure compression and unwanted effects.
15-22 mm carotid / >=15 mm pass mark	Reject universal margin	Individual anatomy and route-specific imaging; de Divitiis supports transclival exposure, not this clearance specification.
15-40 ms pre-emptive body-noise blanking	Unvalidated hypothesis	External EMG/motion/acoustic timing first; false blanking, speech preservation and local failure behavior.

Least burdensome first test

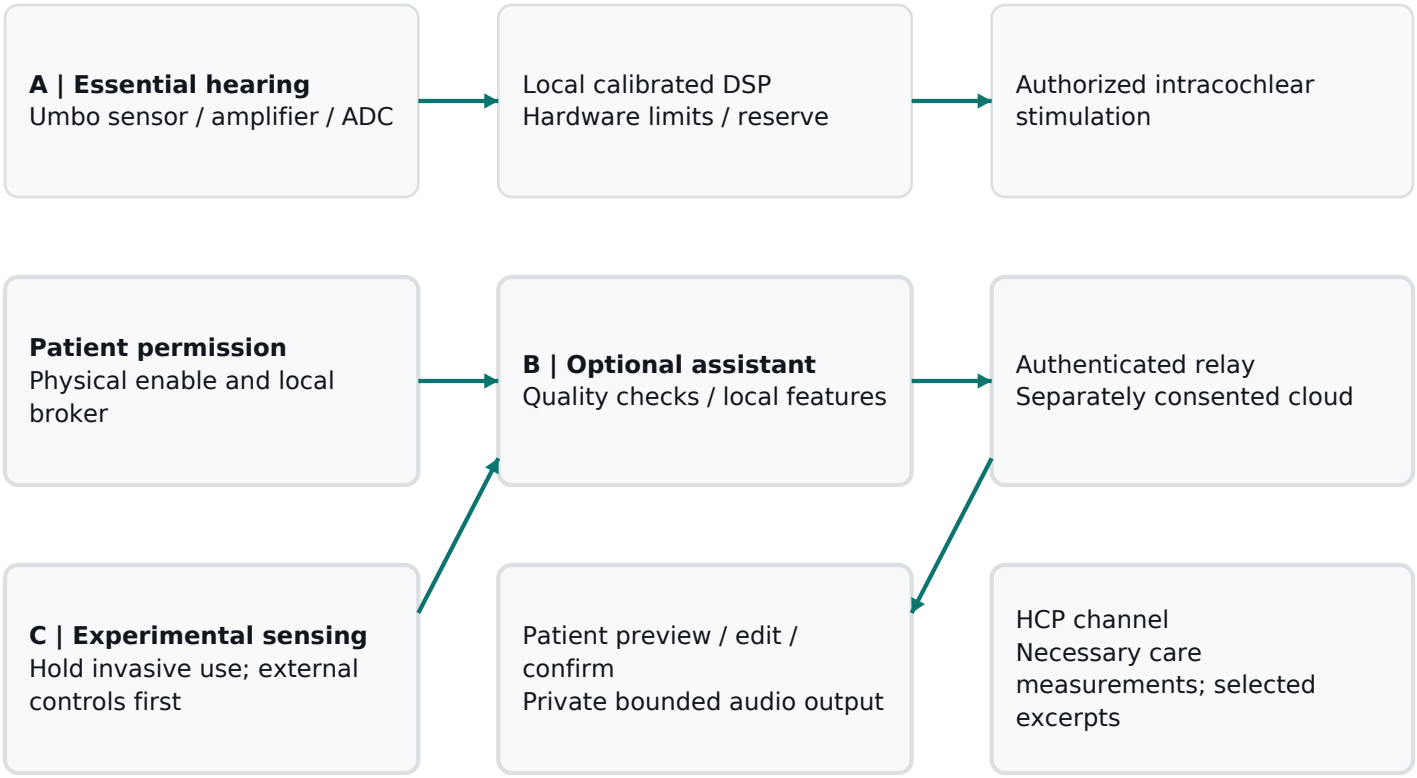
Compare external facial/neck measurements with acoustic, motion, swallow/cough and respiration references. Use unseen utterances, complete held-out days and appropriate participant separation. A new implant must add useful information and patient benefit beyond that comparator.

S1 p.4; S2 pp.3-4,26; Gemini §§1.1-1.3,2.2. R10-R12. Historical evoked tract recordings and clival access are not spontaneous tonsillar semantic recordings. No retrieved chronic target-specific system is a bounded finding, not proof of impossibility.

ARCHITECTURE | PROPOSED

Separate essential hearing from elective assistance

All arrows below are proposed interfaces. Optional software receives bounded input/output permissions; it has no direct authority over electrodes, currents, charge limits or therapeutic stimulation policy.



A/B interface: manufacturer-authorized audio ingress with limiter and deterministic safeguards. C may stop without disabling A. Hazardous essential-core failures may require safe cessation; continuity must not force unsafe stimulation.

Permission	Separate decision
Acquire / decode	Patient enable; no persistent raw/semantic logging by default.
Transfer / persist / export	Purpose, recipient, retention and expiry are explicit. ON is not blanket consent.
Care / stimulation	Clinical authorization does not confer remote ON or model privilege to change medical limits.

S1 pp.5,7; S2 pp.20-21,25. Proposed system requirements; no demonstrated integration.

Choose the signal before the link

A bandwidth limit is not a data-rate guarantee. Select channel count, sample rate and precision from the biological endpoint, then measure useful goodput, power, thermal load and failure behavior.

Scenario arithmetic	Raw / framed rate	Decision limit
16 channels x 1 kS/s x 16 bits	256 / 320 kbit/s	25% overhead is illustrative, not a protocol measurement.
64 channels x 2 kS/s x 16 bits	2.048 / 2.560 Mbit/s	Exceeds LE 2M gross rate; does not prove every Bluetooth PHY is inadequate.
S1: 20 features x 8 bits x 20 Hz	3.2 / 4.0 kbit/s	Lossy candidate; speech information and false-activation effects unvalidated.
Gemini: 16 features x 16 bits x 10 Hz	2.56 / 3.20 kbit/s	Correct arithmetic, not "optimal/adopted." A 10 Hz frame is 100 ms.

Verified envelope	Boundary
US MedRadio 402-405 MHz	25 uW M-EIRP; 300 kHz session bandwidth. Voice communication prohibited. Radiated power is not DC power or charging capability. [R13]
ETSI EN 301 839	ERP and EIRP differ; use the applicable monitoring/channel rules. Do not invent universal 100-250 kbit/s ceilings. [R14]
BLE / Auracast	LE 2M is a specific PHY; optional HDT up to 7.5 Mbit/s was announced before cutoff. Implant support remains unproven. Auracast is broadcast audio, not neural telemetry. [R15]

> Reject Gemini's <6-hour runtime and fixed 25-35 dB tissue-loss claims without an actual battery, antenna and link budget. Sampling at 1-2 kS/s cannot preserve an unfiltered 2 kHz signal band.

S1 p.6; S2 p.14; Gemini §2.3. Test antenna detuning, body-channel distributions, packet loss, charge coupling, sensor/AFE/ADC/DSP/radio/stimulator currents and maximum-load heating.

OFF is a tested state transition

Use a local deterministic permission broker outside the generative model. Patient control must be demonstrated against delayed data, stale credentials and compromised optional components, rather than asserted from a prompt or an API policy.

Failure / control	Required behavior	Verification
Patient OFF	Stop elective acquisition, egress, output and new memory; cancel queues; invalidate the session. Only patient action re-enables.	Logic-analyzer and egress traces; late packets, retry, reconnect and remote requests.
Cloud / phone loss	Optional output stops safely; essential hearing continues within its safety case.	Timeouts, malformed responses, relay loss and maximum adjunct load.
Adjunct battery depletion	Separately enforced essential reserve / power domain.	Rail faults and current limits; no immunity to exhausted essential battery.
Software compromise	Unsigned/replayed updates rejected; known-good recovery; no model access to medical limits.	Secure boot, signed update, rollback and privilege-boundary tests.
Past disclosure	Future transmission revoked; delivered data cannot be recalled by a local switch.	Contractual retention, deletion, access logs and data-flow inventory.

HCP and remote application access

Default care access is necessary integrity and approved-care measurements. Any semantic excerpt needs patient selection of content, purpose, recipient and duration. Screen/window access, typing, private audio and memory are distinct grants. Review before external action; default background screenshots and silent labels off.

S1 pp.5,7; S2 pp.20,25; Gemini §§1.2,2.4. R16-R18. Shutdown timing must be hazard-derived and measured; <10 ms is not an achieved result or a universal legal threshold.

A quality failure is not a diagnosis

Prevent unsupported psychiatric labels without replacing them with unsupported device diagnoses. Nonspecific behavior can have several explanations; passing or failing a sensor checklist neither establishes nor excludes a coexisting clinical condition.

Input / observation	Safe product behavior	Rejected interpretation
Impedance drift, poor SNR, movement	Flag data quality and abstain from automated interpretation; retain calibration/version.	"DEVICE_OR_OTOLOGICAL_STATE" as a proven diagnostic cause.
Pauses, flat prosody, fatigue	Offer patient-controlled communication support; permit direct self-report and clinician review.	Automatic psychiatric diagnosis or automatic attribution to effusion/listening fatigue.
Incoherent generated text	Separate decoder output and language-model transformation; preview, reject and abstain.	Text errors as thought disorder, or posterior 0.92 as a universally valid threshold.
Arousal, sleep difficulty or tinnitus distress	Appropriate clinical assessment independent of optional sensor status.	Tinnitus rebound / overstimulation as the "true" cause; autonomous intracochlear masking.

Provenance class	What it means
Measurement	Sensor, units, calibration, timestamp, quality and collection permission.
Algorithmic proxy	Model/version, defined uncertainty, false-positive behavior and input references; not mood/intent/diagnostic fact.
Patient-approved wording	Approved self-report and edit history; not independent proof of decoder correctness.

Choose S1 p.7 and S2 pp.20-21 over Gemini §2.4. R19-R21. NICE NG155 supports specified psychological stepped care, not this implant or a generic relaxation-audio protocol. No platform-specific psychiatric misdiagnosis incidence is established.

NEW S2 ADDITION | LISTENING SUPPORT

Attention feedback earns a separate experiment

Retain AttentivU as an adjacent external research rationale. Prioritize patient-selected replay, pause or clarification during bounded listening tasks; do not convert a classifier score into a concentration diagnosis or ADHD treatment claim.

Evidence	Finding	Limit
2019 learning study [R22]	36 analyzed adults in video tasks; 12 in preliminary lecture deployment; contingent, random and no-feedback conditions.	Immediate comprehension, not durable learning or clinical efficacy. Historical exception to 2020+ window.
2022/2023 auditory pilot [R23]	14 users; two ear-separated stories; mean attended-stream classification 74% low-channel vs 79% research EEG.	Primary abstract inspected. Not transcription, neural speech reconstruction or CI-user benefit.
Other glasses/workload reports	Useful acquisition and human-factors leads in S2.	Heterogeneous tasks and repeated cohorts are not independent clinical replication.

Three release tests

Test	Evidence needed
Measurement validity	Fix the sensor/algorithm; independent users and sessions; chewing, speech, eye movement and CI-stimulation artifact controls.
Controlled benefit	Contingent cues versus matched/yoked cues and standard support. Measure comprehension, missed information, effort and interruptions separately from the model score.
Repeated use	Drift, voluntary uptake, cue burden and preference. HCP authorization for the scoped clinical function plus patient final control.

> Attention, ADHD and CBT revenue remains zero in this research branch. Hardware approval does not validate a task-state estimator. Scope new trial costs before funding the branch.

Adopt S2 pp.21,23-24. R22-R23; full auditory validation splits and long-term outcomes remain diligence items. No necessary dependency on an invasive Layer C is established.

NEW S2 ADDITION | CALIBRATION

Separate training, context and the release test

Voluntary practice is a bounded experiment, not a two-year guarantee. Training labels come from the participant or prespecified task; model suggestions, screenshots and hidden context must not become circular ground truth.

Stage	Required control	Stop / release rule
1. Define task	Distinguish acoustic, attempted, silently articulated, instructed inner speech and imagery. Fix sensor and task markers.	No pooled performance number across tasks or modalities.
2. Practice voluntarily	Short agreed blocks, breaks and immediate OFF. Personal comfort and clinical limits bound the session.	No covert provocation or training until a score passes.
3. Lock evaluation	Split before overlapping windows; unseen utterances/sessions/participants as required; no release-test tuning.	Neural-only, context-only, declared fusion and no-intent controls.
4. Review output	Word/phoneme errors, consequential errors, false activations, coverage/abstention, drift and latency tails.	Patient acceptance and clinical review where applicable supplement, never waive, safe limits.
5. Adapt cautiously	Versioned offline change, regression, rollback and consent expiry; preserve native-rate archives.	Narrow or stop when the signal or patient benefit fails.

Imagery and pretrained priors

An imagined-person label is a separate hypothesis, not identity proof or recovered private thought. Test unseen targets, non-target imagery and prompt leakage. A pretrained articulatory prior can initialize research but does not validate a new sensor site or eliminate individual calibration.

Adopt S2 pp.25-26; S1 pp.3-4,17. R04-R10. Read-only acquisition and confirmed output precede any extension. Acoustic recognition, neural decoding and language correction retain separate provenance.

Authorization follows the actual function

Define each device configuration, accessory, indication and software function with the relevant sponsor and authorities. Clinical use, manufacturing conformity, radio permission and reimbursement are distinct decisions; none is supplied by a diagram or HCP involvement alone.

Function / trigger	Required diligence	Do not infer
New hearing integration	Partner interface and change pathway; device-specific safety/performance; quality system and clinical evidence.	An approved CI makes a new microphone or electrode route approved.
Significant-risk human investigation	Applicable FDA/IRB authorization and monitoring; route-specific risk-benefit and stopping plan.	A planned 30-day review is guaranteed approval or permission for chronic implantation.
AI changes / PCCP	Describe permitted modifications, validation and implementation controls for the actual authorized function.	A PCCP authorizes a new invasive interface or unrelated therapy.
Listening / concentration support	Function-specific classification; assess invasive/implant modifications separately from external low-risk features.	General-wellness policy exempts the entire implanted platform.
EU / other markets	Device classification, clinical evidence and local radio/accessory obligations; country-specific pathway review.	US or EU status automatically authorizes every geography.

Clinical design remains an investigator decision

Set eligibility, sample size, endpoints, noninferiority/superiority margins and follow-up from the intended population and risk case. Conventional CI eligibility alone does not justify elective pharyngeal implantation. A zero-event series of five to ten participants cannot exclude uncommon severe harm.

S1 p.8; S2 pp.4,21,24; Gemini §§1.4,3.4. R24-R27. QMSR effective 2 February 2026; compliance must be demonstrated. CMS coverage is not market authorization or automatic assistant reimbursement.

A collaboration is a contract, not a premise

The scientific option has value only if the parties can deliver the exact rights, authorized interface and funded support duties. Corporate relationships, research provenance and public repositories do not establish a licensable commercial bundle.

Workstream	Evidence required	Decision impact
Device / platform partner	Authorized configuration, interface documentation, sponsor authority, manufacturing and safety responsibilities.	No exclusive commitment without verified access and counterparties.
Background / foreground IP	Patent and software inventory; licence scope, sublicensing, improvements, data and exit rights.	Novelty and freedom to operate remain distinct from scientific feasibility.
Research software	Version-specific code/model/data licences; dependencies and distribution/use model.	Copyleft is not automatically a blanket ban; unvalidated psychiatric labels and background memory require removal or containment.
Cloud / voice services	Service/region/retention, model migration, security, voice consent and rights, support SLA and failure fallback.	No guaranteed Azure/OpenAI feature parity, permanent price or seven-year model availability.
Long service cohort	Calibration, drift, replacements, recall/explant and follow-up funding; data portability and vendor exit.	Seven years is a planning cohort cap, not demonstrated life or a required therapy duration.

> Auracast is a broadcast input option. Shared-code encryption is not individual medical authorization; default assistant output to a private authenticated route. No public third-party implant-control right has been established.

S1 pp.9-14; S2 pp.13,20-25; engineering review. R16-R18,R28-R30. Retain GPL and commercial-licence questions for counsel; reject Gemini's categorical proprietary-weight "contamination" conclusion.

Differentiation must survive comparison

The investable claim is a measurable improvement for a defined patient and care pathway. Invisibility, institutional affiliation, a generic AI stack or component sensitivity alone does not establish unique clinical value.

Buyer / user	Value hypothesis	Independent proof
Eligible CI recipient	Less external hardware burden with useful reliable hearing.	Speech in noise, self/body noise, comfort, daily use, failures and surgery/revision burden versus current options.
Assistant user	Easier clarification, drafting and controlled private output.	Task completion, correction burden, unwanted output, listening effort and sustained preference versus phone/native CI support.
Clinician / centre	Better troubleshooting with less avoidable work.	Attribution quality, alert yield, calibration time, escalation workload and traceable measurement provenance.
Payer / provider / OEM	Benefit commensurate with total cost and integration burden.	Coverage fit, actual-price uptake, manufacturing/service costs, authorised interfaces, liability and maintenance obligations.

Competitive status is product-specific

Comparator	Verified scope	Implication
MED-EL TICI UNICO [R30]	Manufacturer European CE-mark launch announcement, September 2026; not a US approval claim.	No defensible first-invisible-implant claim for this proposal.
Envoy Acclaim [R31]	Second modular PMA submission announced October 2026; investigational.	Submission is not approval, equivalence or available revenue.
Cochlear investigational TICI [R32]	Small human feasibility study; microphone-ingress failures described.	Integrated durability and safe fallback require proof beyond bench sensor performance.

S1 p.4; S2 pp.2,4,21; engineering_market_review.md. Candidate benefits are hypotheses. Generic drafting does not automatically require a prescription; clinical functions have separate authorization and evidence gates.

Three different investments, three funding burdens

The merged package uses S2 as the primary model because its uploaded annual schedules are now reconciled. S1 remains a complete alternative, not a numerical correction of S2. M004 assumes the central hardware hurdle away; it cannot validate it.

S2 boundary	10y NPV / peak cash	What the investor funds
Whole programme - base core	-$95.53M / $197.15M	New hearing platform; base does not count paid assistant revenue.
Whole programme - best integrated	$13.99M / $143.04M	Conditional earlier hardware success plus assistant revenue; stronger assumptions.
Partner build - develop software	-$10.44M / $22.03M	Sponsor funds hardware; developer still funds decoder/platform development.
Licensed software - base	-$2.05M / $8.63M	Sponsor-approved platform and licensed software availability; narrower developer duties.
Licensed software - upside	$7.91M / $6.95M	$1,800 fee / 90% attachment; unobserved demand case.

Accounting convention	Locked interpretation
Time / prices	Real 2026 USD; end-2026 valuation; 2027-2036 year-end cash flows; 15% real discount; zero terminal value.
Units	Hardware sales count implanted ears. Subscribers, year-end users and billed patient-years are different measures.
Ownership / funding	Sponsor/developer contribution shares remain to be negotiated. NPVs are project values, not equity prices.

S2: Forecast, Valuation, Partner option and Platform option tabs; financial_comparison.json. Do not add any boundaries or infer success probabilities from scenario labels.

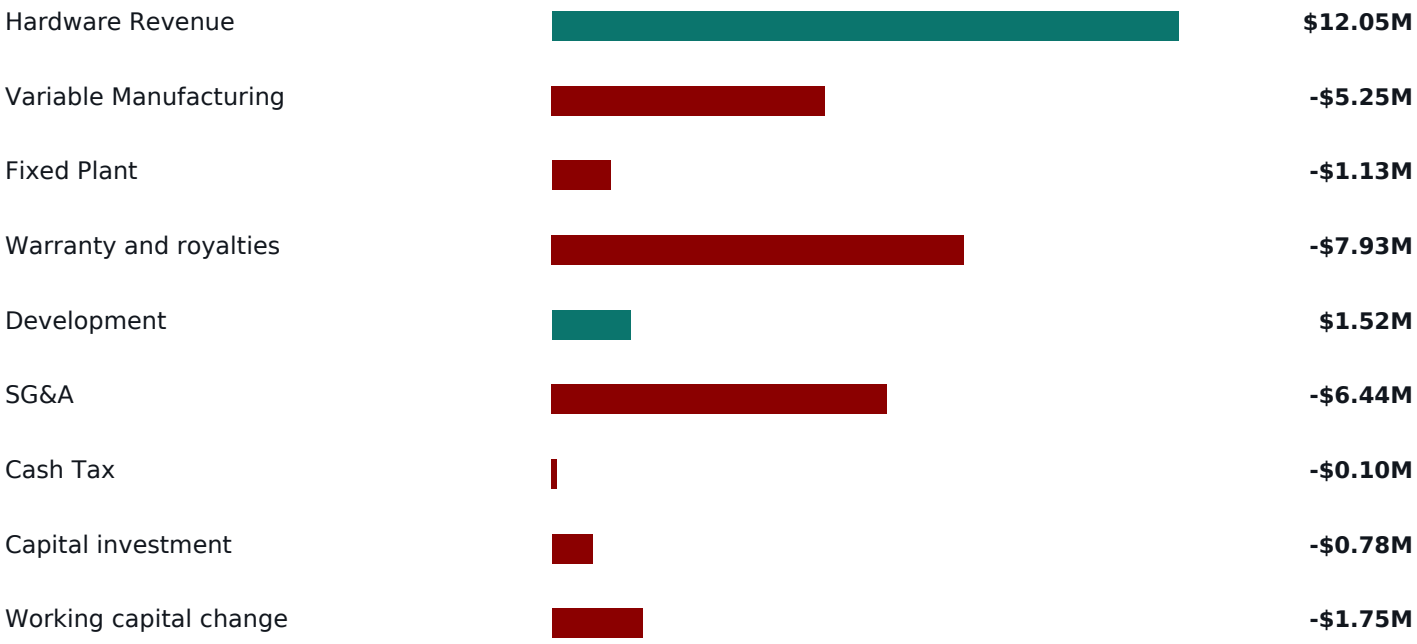
RECONCILIATION | WHOLE PROGRAMME

Higher sales do not guarantee higher NPV

S2 base NPV is $9.82M below S1. The difference is fully explained by annual cash-flow components, including additional warranty/royalty and commercial costs; matching the development and capital totals does not make the schedules equivalent.

Measure	S1 alternative	S2 primary
Base NPV at 15%	-$85.71M	-$95.53M
2036 implanted ears	6,000	8,800
2036 revenue	$121.20M	$168.05M
Peak cash	$186.08M	$197.15M
Development / capital	$146M / $35M	$146M / $35M

Discounted contribution to S2 minus S1 NPV



Exact additive bridge from uploaded annuals; residual below $0.000001. Tax: S1 21%, S2 25%; S2 base real ASP decline 1.5%, warranty 3%, royalty 5%. Both base models are materially negative. Financial audit: 93/93 original-cache checks pass.

Separate billing timing from installed-base ramps

Both S1 and S2 bill new subscribers for half a year. Their 2036 software revenue differs because the annual eligible hardware ramps differ; it is not evidence that one model bills year-end users incorrectly.

Licensed software base	S1 alternative	S2 primary
2036 billed patient-years	15,185.45	17,461.03
2036 year-end users	18,837.62	22,000.16
2036 revenue at $1,200	$18.22M	$20.95M
10-year NPV	-$1.59M	-$2.05M
Peak cash	$8.00M	$8.63M
Tax / software working capital	21% / 5%	25% / 12%

S2 partner build versus licensed software

Responsibility	Partner development	Licensed software
Development total	$22.50M	$8.25M
Software capital total	$2.75M	$0.65M
10-year NPV	-$10.44M	-$2.05M

> Licensed software improves NPV by $8.39M relative to partner development because the assumed responsibilities differ. The comparison requires an executed licence and approved platform; it is not a free reduction in biological or development risk.

S1:M004 and S2:Partner option / Platform option. Default S2 fee $1,200; attachment 60%; retention 88%; upfront rights $0.50M; $60 licence and $300 support per billed patient-year plus 10% sponsor share.

Close the ledger before adopting the headline

Some Gemini rounded free-cash-flow rows reproduce S2 headline NPVs while their operating assumptions and software totals do not reconcile. Retain the actual S2 workbook schedules and disclose Gemini discrepancies instead of averaging the streams.

Gemini location / assertion	Recalculation / conflict	Merge disposition
§5.1 Tab1 base ramp	4,200 ears / $88.8M in 2036 differs from S2's 8,800 / $168.05M; similar NPV alone does not validate drivers.	Reject as replacement annuals; use S2 Forecast / Markets.
§5.1 Tab2 base cash flows	Stated $8.6M peak versus $10.0M maximum cumulative deficit from the listed FCF.	Use reconciled S2 $8.63M; preserve discrepancy in audit.
§5.1 Tab2 upside cash flows	Listed FCF imply about $11.06M NPV at 15%, not $7.9M.	Use S2 $7.91M; no unexplained plug.
§1.1 / Tab2 17,500 subscribers	S2 base has about 17,461 billed patient-years and 22,000 year-end users in 2036.	Label denominators explicitly; do not relabel billed years as people.
§1.4 later capital ceilings	Specific release amounts and fixed clinical milestones lack authorization and validated protocols.	Only initial up-to-$1.5M diligence decision; later spending needs new approval.

A single return definition

Project cash return = sum of annual free cash flow / absolute sum of negative annual free cash flow. It differs from NPV, IRR, equity return and development-plus-capital return. Apply it consistently; never mix denominators in an executive comparison.

Source: financial_comparison.json, S3 CSV reconstruction and S2 original-cache reconciliation. Scenario NPVs remain conditional and unweighted. A positive upside does not establish expected value or willingness to pay.

Model the runway, not only the terminal year

Annual free cash flow below is reproduced from the reconciled S2 schedules. Negative values require funding. The licensed-software case does not carry the sponsor's hardware cash burn and cannot be added to the whole-program case.

Year	Core base	Integrated best	Partner develop	Licensed base
2027	-12.00	-16.00	-3.75	-1.60
2028	-18.00	-24.00	-6.00	-1.75
2029	-26.00	-36.00	-5.50	-1.60
2030	-34.00	-38.02	-2.66	-1.47
2031	-36.00	-24.91	-2.33	-1.32
2032	-31.78	-4.11	-1.59	-0.89
2033	-25.30	30.08	-0.20	0.11
2034	-14.07	74.12	1.93	1.83
2035	6.13	108.69	5.11	4.51
2036	36.65	158.55	9.44	6.76

Annual FCF in real 2026 USD millions. S2 Forecast!D42:M42; Partner option!D38:M38; Platform option!D53:M53 and integrated scenario reconstruction.

Downside is a scenario, not a loss cap

Exposure	Practical release condition
Delay and failed feasibility	S2 shifts sales/factory costs without extra waiting burn; prolonged staff/trial/quality costs need a separate funded-delay stress.
Service / retention burden	Actual-price uptake, cancellation reasons, clinical support, voice use and migration costs measured before launch.
Unpriced risk	A seven-year cohort cap does not fund post-2036 duties. Specify run-off costs and revenue; S1 support reserve is a separate assumption-led stress, not a verified liability.

PLANNING STRESS | SEPARATE FROM BASE

Delay and surviving service duties consume cash

These illustrative S2 stresses test additional obligations and timing risk. They are separate cases, not established contractual liabilities or a probability-weighted forecast. Do not add the support and delay stresses without an explicit combined scenario.

Unfunded seven-year support run-off

Licensed case	Reserve value at 2036	Cost PV at 2026	Adjusted NPV15
Base	$22.706M	$5.613M	-$7.665M
Upside	$34.059M	$8.419M	-$0.504M

Assume no future renewal revenue and a noncancellable $300 annual service obligation for the cohorts surviving at end-2036. Hold those survivors fixed and fund their remaining seven-year terms, including the final half-year. Discount future service costs at 15%; no incremental tax benefit is credited to the reserve.

Launch delay with additional annual cash burn

Delay	Core base NPV15	Licensed base NPV15	Licensed peak cash
1 year	-$119.191M	-$5.668M	$11.954M
2 years	-$132.440M	-$8.282M	$15.646M

Core: apply the original S2 launch-delay scenario and add $4M cash burn per delayed year, with no incremental tax shield. Licensed software: shift the eligible sales/cohort ramp and add $1M per delayed year; retain the model tax and working-capital calculation. No terminal recovery is assumed.

> The S2 upside changes from +$7.914M to -$0.504M under this unfunded-support stress. Specify contractual service obligations, renewal rights, termination and financing before relying on an attractive software scenario.

Source: financial_comparison.json, planning_stresses; S2 cohort-specific recomputation. The earlier S1 support stress remains a separate alternative in the archived S1 model; its different cohort ramp is not transplanted into S2. These stress values do not authorize additional capital.

Buy decisive evidence in the first 90 days

Approve a ceiling, not a requirement to spend it. The sponsor/developer contribution split and work orders remain to be negotiated. Each release must identify an owner, protocol, source data and a clear stop condition.

Workstream	Ceiling	Required deliverable
Signal evidence	$400k	Exact task and target subgroup; pre-registered signal-origin and external-comparator protocol.
Hardware and power	$350k	Assembly/interface access; anatomy-specific hazards; measured power/latency and OFF prototype.
Regulation, privacy and IP	$250k	Intended-use split, agency-meeting package, privacy data flows, rights and liability review.
Patients, centres and payers	$200k	Actual-price value tests, centre capacity and current-care alternatives; no assumed paid demand.
Replication and security	$150k	Independent reproduction and fault/egress traces; leakage and version controls.
Contingency	$150k	Release only for a documented evidence gap; unused funds stay uncommitted.

> **Total: $1.50M.** Diligence authorization does not authorize chronic human implantation or commercial deployment.

Later period	New decision evidence
Months 3-9	Replicated signal or a justified external-only scope; measured patient controls and initial comparative benefit
Months 9-18	Authorized preclinical program where warranted; packaging, reliability, cost and service evidence
Months 18-24	Integrated fault tests, regulator feedback, executed rights and funded clinical/support obligations

editorial_controls.json. S1 p.17 workstream budget retained; later numerical ceilings superseded. S2 p.4 gates and Gemini candidate tests contribute only where protocol justified.

A green light must name its boundaries

Advance only the function that passes its evidence and responsibility gates. A failed invasive branch may stop while justified external assistance or microphone partnership work continues. Calendar progress and a persuasive demo are not release evidence.

Dimension	Pass evidence	Current state
Biological information	Independent signal-only performance and incremental benefit over a practical external comparator.	Open for proposed pharyngeal route.
Patient benefit	Prespecified patient-relevant endpoints, representative users, controlled comparison and tolerable burden.	Unestablished for new combined system.
Physical safety	Validated anatomy, fixation/barrier, power/thermal limits, fault containment and chronic plan.	Component evidence only; system work required.
Control and privacy	No unauthorized egress/actions; OFF and retention behavior verified; no automatic diagnostic claims.	Requirements specified, implementation not proven.
Rights and approval	Executable licences, sponsor interfaces, appropriate regulatory/clinical pathway and funded duties.	Unverified counterparties and approvals.
Economic viability	Reconciled schedules plus observed price/uptake, service costs, financing and termination obligations.	Arithmetic verified; market assumptions remain open.

Delivery-excellence rubric

Criterion	Pass rule
Traceable	Every decision claim has a source, input location or explicit assumption; papers deduplicated by DOI.
Testable	Acceptance limits and statistical units set before independent testing; no arbitrary copied threshold.
Usable	Decision, owner and stop rule visible; model boundaries never added; final document visually checked.

Scientific/engineering/regulatory reviewer synthesis. This is desk diligence, not an external clinical panel, regulator determination, financial audit or proof of feasibility.

Primary sources 1-8

Selected decision references only. The workbook archive retains the full S1 ledger and 141 source-attributed S2 bibliography entries. Original codes are namespaced; repeated reports and participants are not new replication.

Canonical source	Verified access / input mapping
R01 Yeiser et al. The UmboMic: A PVDF Cantilever Microphone (2024)	https://doi.org/10.1088/1361-6439/ad5c6d Primary abstract/figures; initial component evidence. **Map:** S1:SCL001; S2:SCL001; Gemini:SCL006
R02 Wawrzynek et al. UmboMic characterization and testing (2026)	https://doi.org/10.1007/s10162-026-01039-x Indexed primary methods/results/discussion; publisher abstract. **Map:** S1:SCL002; S2:SCL002; Gemini:SCL007
R03 Comens et al. Cadaveric sheep implantation/fixation (2026)	https://doi.org/10.1007/s10162-026-01056-w Primary full text. Name-only correction: 10.1007/s10162-026-01072-w. **Map:** S1:SCL003; S2:SCL003-004; Gemini:SCL008
R04 Willett et al. A high-performance speech neuroprosthesis (2023)	https://doi.org/10.1038/s41586-023-06377-x Primary paper; task/population limits retained. **Map:** S1:SCL004; Gemini:SCL001
R05 Metzger et al. Speech decoding and avatar control (2023)	https://doi.org/10.1038/s41586-023-06443-4 Primary paper; surface ECoG. **Map:** S1:SCL005; Gemini:SCL002
R06 Card et al. Accurate rapidly calibrating speech neuroprosthesis (2024)	https://doi.org/10.1056/NEJMoa2314132 Primary full-text study in S1; selected labeled conversation denominator. **Map:** S1:SCL006; S2:SCL009; Gemini:SCL003
R07 Wairagkar et al. Instantaneous voice-synthesis neuroprosthesis (2025)	https://doi.org/10.1038/s41586-025-09127-3 Primary paper; same T15 family as R06. **Map:** S1:SCL007; S2:SCL010; Gemini:SCL004
R08 Kunz et al. Inner speech in motor cortex (2025)	https://doi.org/10.1016/j.cell.2025.06.015 Primary paper; task-dependent decoding and gating. **Map:** S1:SCL008; S2:SCL011; Gemini:SCL005

Primary sources 9-16

Selected decision references only. The workbook archive retains the full S1 ledger and 141 source-attributed S2 bibliography entries. Original codes are namespaced; repeated reports and participants are not new replication.

Canonical source	Verified access / input mapping
R09 Littlejohn et al. Streaming brain-to-voice (2025)	https://doi.org/10.1038/s41593-025-01905-6 Indexed primary Extended Data Fig.3; general-set latency. **Map:** S2:SD003
R10 Gaddy & Klein. Digital Voicing of Silent Speech (2020)	https://doi.org/10.18653/v1/2020.emnlp-main.445 Primary abstract/paper; silently articulated facial EMG. **Map:** S1:SCL009
R11 de Divitiis et al. Transoral-transclival anatomy (2004)	https://doi.org/10.1227/01.neu.0000097271.55741.60 Primary abstract; historical exception; no universal cuff clearance. **Map:** S1:SCL013; S2:SCL008; Gemini:SCL011
R12 Sahan & Muluk. Tonsillar fossa-carotid distance in children (2020)	https://doi.org/10.1016/j.ijporl.2020.110209 Primary abstract; pediatric anatomy, not adult complication rate. **Map:** S1:ST001
R13 US FCC. MedRadio radiated-power / voice and bandwidth rules	https://www.govinfo.gov/content/pkg/CFR-2025-title47-vol5/pdf/CFR-2025-title47-vol5-sec95-2565.pdf Official 2025 CFR shared page includes 95.2567. Voice/bandwidth: 95.2571/2573. **Map:** S1:ST007-008; S2:ST004
R14 ETSI. EN 301 839 V2.1.1 (2016)	https://www.etsi.org/deliver/etsi_en/301800_301899/301839/02.01.01_60/en_301839v020101p.pdf Official standard; verify applicable conformity version at freeze. **Map:** S1:ST009; S2:ST005
R15 Bluetooth SIG. High Data Throughput technical overview (3 Nov 2026)	https://www.bluetooth.com/bluetooth-high-data-throughput-technical-overview/ Official technical announcement; not implant availability. **Map:** S1:ST011; author Alfredo Perez
R16 FDA. Device cybersecurity guidance (2026)	https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cybersecurity-medical-devices-quality-management-system-considerations-and-content-premarket Official guidance; exact-system implementation still needed. **Map:** S1:ST013

Primary sources 17-24

Selected decision references only. The workbook archive retains the full S1 ledger and 141 source-attributed S2 bibliography entries. Original codes are namespaced; repeated reports and participants are not new replication.

Canonical source	Verified access / input mapping
R17 FDA. Percept aDBS P960009/S478 SSED (2025)	https://www.accessdata.fda.gov/cdrh_docs/pdf/P960009S478B.pdf Official device-specific control analogy; not speech/tonsillar authorization. **Map:** S1:SRM015; S2:ST001/BC001; Gemini:SRM002
R18 HL7. FHIR R4 AuditEvent and Provenance	https://hl7.org/fhir/R4/auditevent.html Official Provenance / AuditEvent documentation; schema does not enforce immutability. **Map:** S1:SA003-004
R19 NIST. Generative AI profile, AI 600-1 (2024)	https://doi.org/10.6028/NIST.AI.600-1 Official risk framework; not device-specific psychiatric incidence. **Map:** S1:SA001
R20 WHO. Large multimodal model governance (2024)	https://www.who.int/news/item/18-01-2024-who-releases-ai-ethics-and-governance-guidance-for-large-multi-modal-models Official guidance; general automation/privacy risks. **Map:** S1:SA002
R21 NICE. NG155 tinnitus recommendations (2020)	https://www.nice.org.uk/guidance/ng155/chapter/Recommendations Official recommendations; specific psychological care; no implant endorsement. **Map:** S1:SRM017; S2:SCL013; Gemini:SRM013
R22 Kosmyna & Maes. AttentivU personalized learning (2019)	https://doi.org/10.3390/s19235200 Indexed primary methods/results; historical exception; immediate outcomes. **Map:** S2:AT001
R23 Kosmyna, Balyan & Hauptmann. Target Speaker Detection (2022/2023)	https://doi.org/10.1007/978-3-031-18458-1_30 Publisher abstract only; full validation splits not verified. **Map:** S2:AT012
R24 FDA. Quality Management System Regulation	https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr Official; effective 2 February 2026. **Map:** S1:SRM001; S2:SRM006; Gemini:SRM003

Primary sources 25-32

Selected decision references only. The workbook archive retains the full S1 ledger and 141 source-attributed S2 bibliography entries. Original codes are namespaced; repeated reports and participants are not new replication.

Canonical source	Verified access / input mapping
R25 FDA. AI-enabled device PCCP guidance (2025)	https://www.fda.gov/media/166704/download Official; bounded changes, not a new indication authorization. **Map:** S1:SRM002; S2:SRM008/ST014
R26 FDA. General Wellness: Low Risk Devices (2026)	https://www.fda.gov/media/90652/download Official policy; invasive/implanted platform not blanket-exempt. **Map:** S2:AR004
R27 CMS. Cochlear implantation NCD 50.3	https://www.cms.gov/medicare-coverage-database/view/ncd.aspx?NCDId=245 Official coverage policy; not approval of adjunct or cuff. **Map:** S1:SRM014; S2:SRM004; Gemini:SRM010
R28 Bluetooth SIG. Auracast operation	https://www.bluetooth.com/auracast/how-it-works/ Official broadcast description; individual medical permissions separate. **Map:** S2:SP008
R29 NeuroSkill / NeuroLoop research repositories	https://github.com/NeuroSkill-com/neuroloop Version-specific research code documentation; not clinical efficacy. **Map:** S2:RM007/EX005; Gemini:ST013
R30 MED-EL. TICI UNICO announcement (29 September 2026)	https://www.medel.com/en-us/press-room/press-list/2026/09/29/med-el-launches-tici-unico-world's-first-and-only-totally-invisible-cochlear-implant Official manufacturer announcement; no UmboMic comparative superiority. **Map:** S1:SX001; S2:SCL006; Gemini:SP001
R31 Envoy Medical. Acclaim second modular PMA submission (12 October 2026)	https://www.envoymedical.com/news/envoy-medical-submits-second-module-of-modular-premarket-approval-application-to-fda-for-breakthrough-designated-acclaimr-cochlear-implant Official issuer announcement; submission is not approval. **Map:** S1:SX002; S2:SRM003; Gemini:SP004
R32 Cochlear investigators. Totally implantable CI feasibility (2025)	https://pubmed.ncbi.nlm.nih.gov/41243136/ Primary indexed study; distinct device and small cohort. **Map:** S1:ST014

Validation: 93 original-cache comparisons pass; additive bridge residuals are below $0.000001. This checks arithmetic at supplied assumptions, not clinical success or market uptake. All PDF pages were rendered for visual inspection.